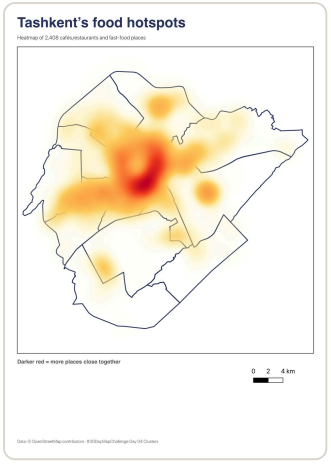
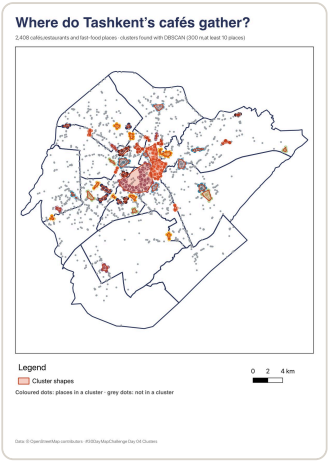
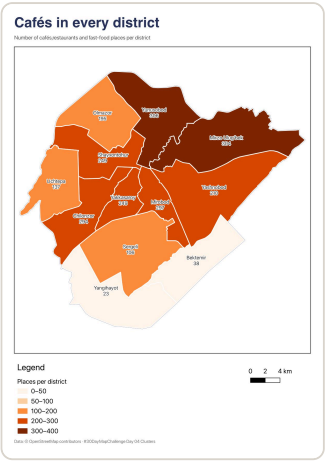
🫧 04-kun · Klasterlar: Toshkent kafelari qayerda to‘planadi?

QGIS bo‘yicha yangi boshlovchilar uchun dars. QGIS mutaxassislar foydalanadigan bepul xarita dasturi. Kod yozish ham, tajriba ham kerak emas.

Siz OpenStreetMap’dagi **2 408 ta kafe, restoran va fastfud joyini** Toshkent xaritasiga joylashtirasiz, **issiqlik xaritasida** ular qayerda to‘planishini ko‘rasiz, **DBSCAN** vositasi yordamida QGIS’ga **klasterlarni** topdirasiz, har bir klaster atrofiga shakl chizasiz, har bir **tumandagi** joylarni sanaysiz va chop etishga tayyor xarita yasaysiz.

⏱️ Taxminan 45 daqiqa (va QGIS’ni o‘rnatish uchun 10 daqiqa) · 💻 Windows, macOS yoki Linux · 🔑 Akkaunt kerak emas

Issiqlik xaritasi	DBSCAN klasterlari	Tumanlar bo‘yicha joylar
		

📚 0-qism · Asosiy tushunchalar

Nuqtalar. Har bir kafe joylashuvi va bir nechta ustuni (`name` , `kind`) bor bitta nuqta.

Klaster. Bir-biriga qolganlariga qaraganda ancha yaqin joylashgan nuqtalar guruhi. Klasterlar hayot qayerda qaynayotganini ko‘rsatadi: savdo ko‘chalari, shahar markazi, bozor.

Issiqlik xaritasi. Har bir nuqta atrofiga ozgina "issiqlik" tarqatadi. Ko‘p nuqta yonma-yon bo‘lgan joyda issiqlik qo‘shilib, qizil rangga aylanadi. Bu klasterlarni *ko‘rishning* tez usuli.

DBSCAN. Klasterlarni ikki oddiy qoida bo‘yicha *topadigan* vosita:

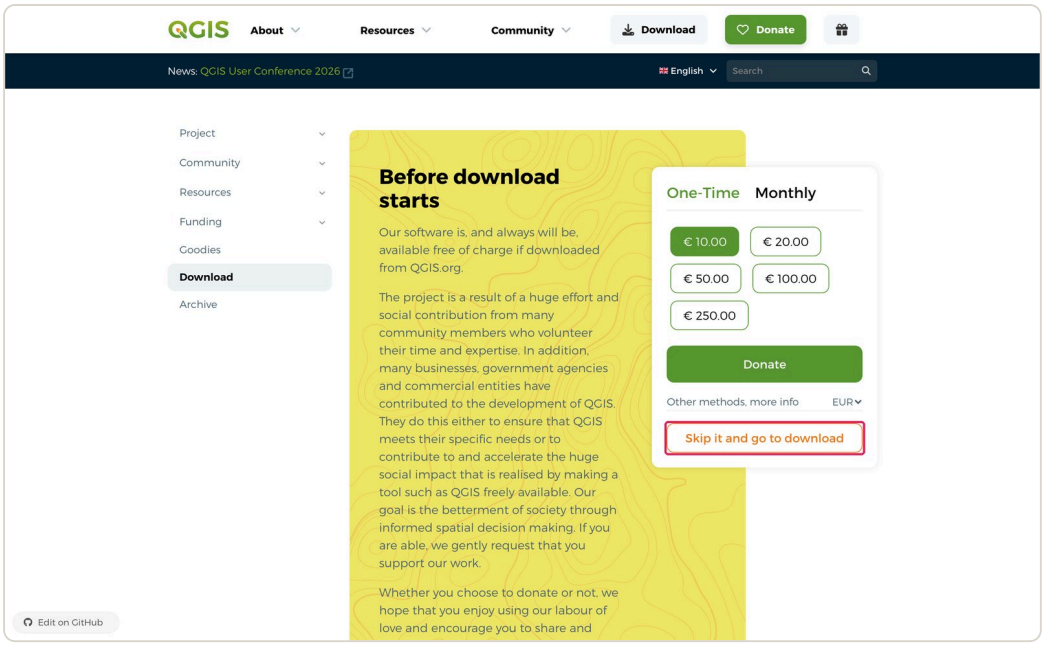
1. **Masofa:** ikki joy orasidagi masofa, masalan, **300 m** dan kam bo‘lsa, ular qo‘shni hisoblanadi.
2. **Eng kichik o‘lcham:** guruh klaster hisoblanishi uchun unda kamida, masalan, **10 ta joy** bo‘lishi kerak.

Hech bir guruhga yaqin bo‘lmagan joylar **shovqin** (noise) deyiladi: ular hech qaysi klasterga kirmaydi.

Proyeksiya. Ma’lumotlarimiz gradusda (uzunlik va kenglik) yozilgan. DBSCAN’ga **metr** kerak, shuning uchun avval nuqtalarni metrli koordinatalar tizimiga o‘tkazamiz.

💾 1-qism · QGIS’ni o‘rnating

1. **qgis.org/download** saytiga kiring. Sayt avval xayriya qilishni taklif qiladi. QGIS bepul: **Skip it and go to download** (O‘tkazib yuborish va yuklab olishga o‘tish) tugmasini bosing.



2. Sayt tizimingizni (Windows yoki macOS) o‘zi aniqlaydi. **Download LTR** (Long Term Release, eng barqaror versiya) tugmasini bosing.



3. Uni boshqa dasturlar kabi o‘rnating: **Windows**’da `.msi` faylini ishga tushiring va oxirigacha *Next* ni bosing. **macOS**’da `.dmg` faylini oching va QGIS’ni *Applications* papkasiga sudrab tashlang.

> 💡 Darsdagi skrinshotlar Mac’dagi QGIS 3.34 da olingan. Yangi versiyalar (3.44 va undan keyingilar) deyarli bir xil ko‘rinadi. Windows’da menyular (**Project**, **View**, **Layer**, **Vector**…) QGIS oynasining yuqori qismida, Mac’da esa ekranning yuqori qismida joylashgan. Darsda tugma va menyu nomlari QGIS’ning inglizcha versiyasidagidek berilgan.

📥 2-qism · Ma’lumotlarni yuklab oling

Ikkala faylni bitta papkaga yuklab oling:

Fayl	Ichida nima bor
`tashkent_food_places.geojson`	2 408 ta kafe, restoran va fastfud joyi (`name` , `kind` ustunlari)
`tashkent_districts.geojson`	Toshkent shahrining 12 ta tumani (`district` ustuni)

⬇️ Yoki ikkalasini birdaniga: **day04_clusters_data.zip**, so‘ng arxivni oching.

GitHub’dan bitta faylni yuklab olish uchun uning nomini, so‘ng o‘ng yuqoridagi **Download raw file** (⬇) tugmasini bosing.

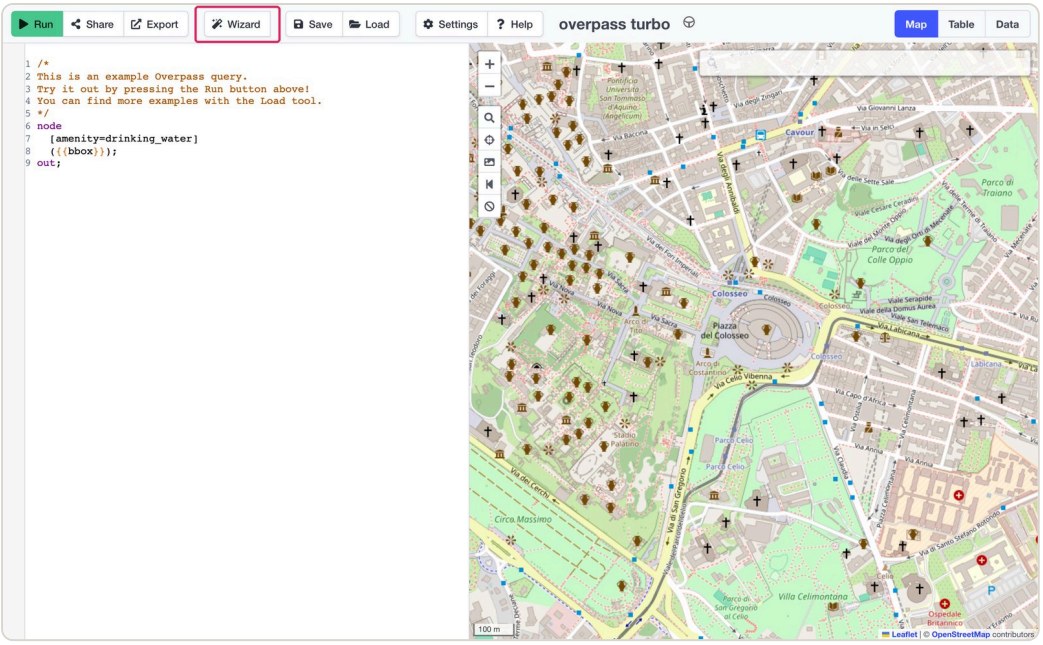
📡 Ma’lumotlar qayerdan olingan? O‘zingiznikini OpenStreetMap’dan yuklab oling

Kafelarimiz **OpenStreetMap** (OSM) dan olingan: bu ko‘ngillilar yaratayotgan bepul dunyo xaritasi. Undan istalgan joylarni o‘zingiz yuklab olishingiz mumkin: maktablar, dorixonalar, masjidlar, avtobus bekatlari, bog‘lar… Buning ikki usuli bor.

	A · Overpass Turbo	B · Geofabrik
Nima uchun qulay	bitta shahar va bir yoki bir necha turdagi joylar	butun mamlakat va birdaniga ko‘plab qatlamlar
Qanday	brauzerda so‘rov yoziladi	bitta katta zip-arxiv yuklab olinadi
Hajmi	kichik (Toshkent kafelari uchun taxminan 1 MB)	katta (O‘zbekiston uchun taxminan 260 MB)

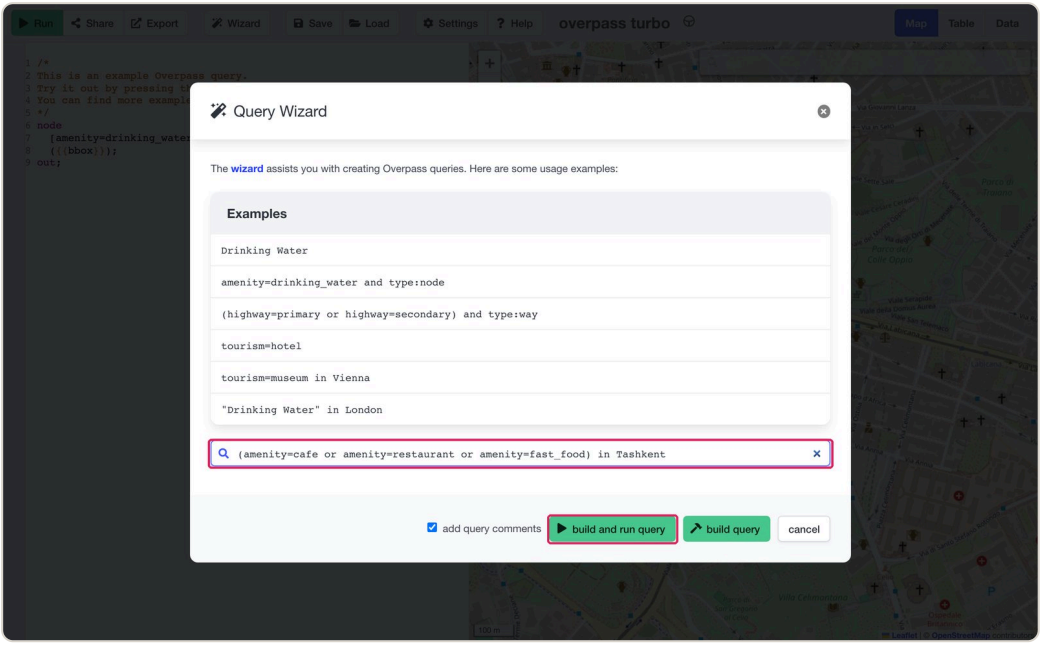
A · Overpass Turbo (brauzerda)

1. **overpass-turbo.eu** ni oching va yuqoridagi **Wizard** (Yordamchi) tugmasini bosing. (Xaritada boshqa shahar ko‘rinsa, xavotir olmang: qidiruv uni o‘zi suradi.)

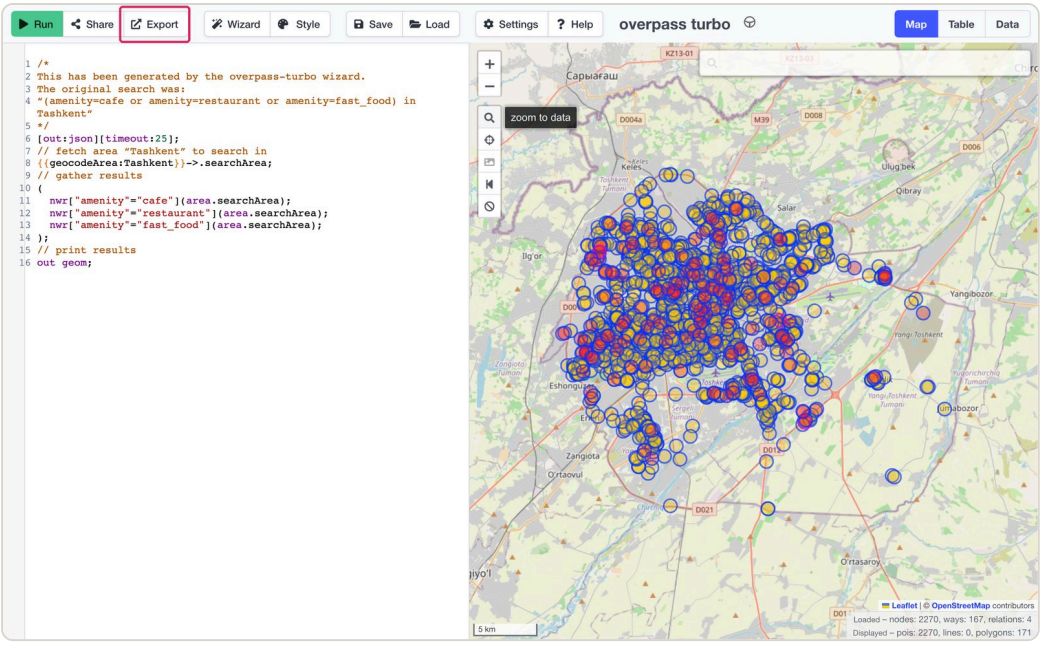


2. Nimani va qayerdan qidirayotganingizni yozing va **build and run query** (so‘rovni tuzish va bajarish) ni bosing:

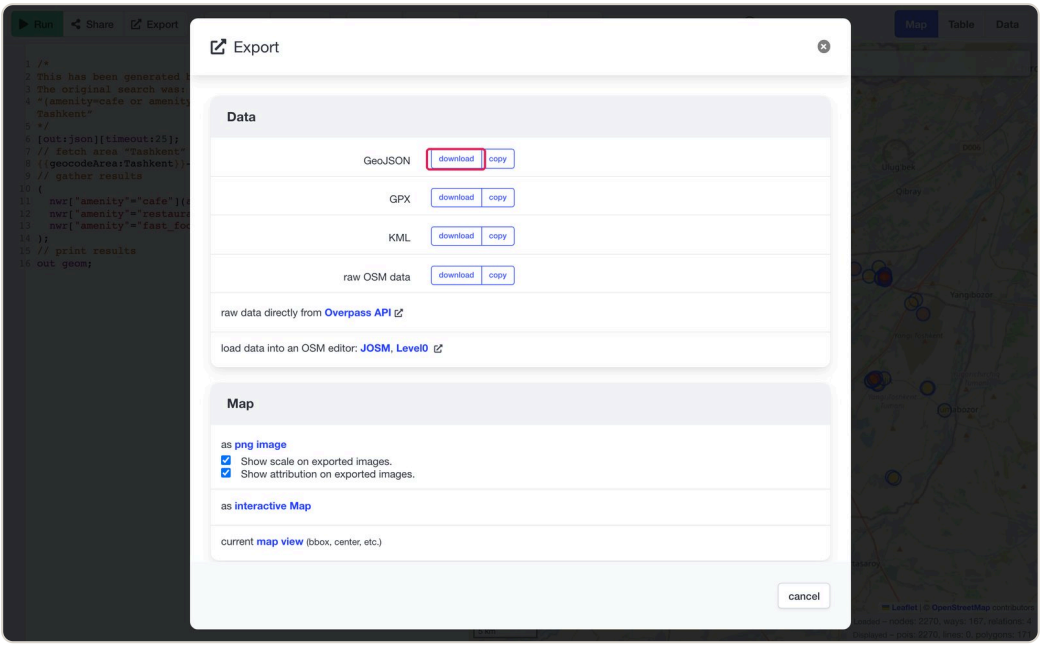
```
(amenity=cafe or amenity=restaurant or amenity=fast_food) in Tashkent
```



3. Bir necha soniya kuting. Joylar xaritada paydo bo‘ladi (ko‘rinmasa, xaritaning chap tomonidagi 🔍 **zoom to data** ni bosing). Chapdagi matn Yordamchi siz uchun **Overpass** tilida yozgan haqiqiy so‘rov.



4. **Export** ni bosing → **GeoJSON** yonidagi **download** ni bosing. `export.geojson` nomli fayl yuklab olinadi.



5. `export.geojson` ni QGIS'ga sudrab tashlang. QGIS qaysi qismini qo'shishni so'raydi, chunki OSM'da ba'zi joylar **nuqta**, ba'zilari esa **bino konturi** (poligon) sifatida saqlanadi. **Point** qatlamini tanlang (yoki ikkalasini tanlab, poligonlarni **Vector** → **Geometry Tools** → **Centroids** orqali nuqtaga aylantiring).

Nima yozish kerak? OSM'da har bir obyektning `kalit=qiymat` ko'rinishidagi **tegi** bor. Bir nechta foydali teglar:

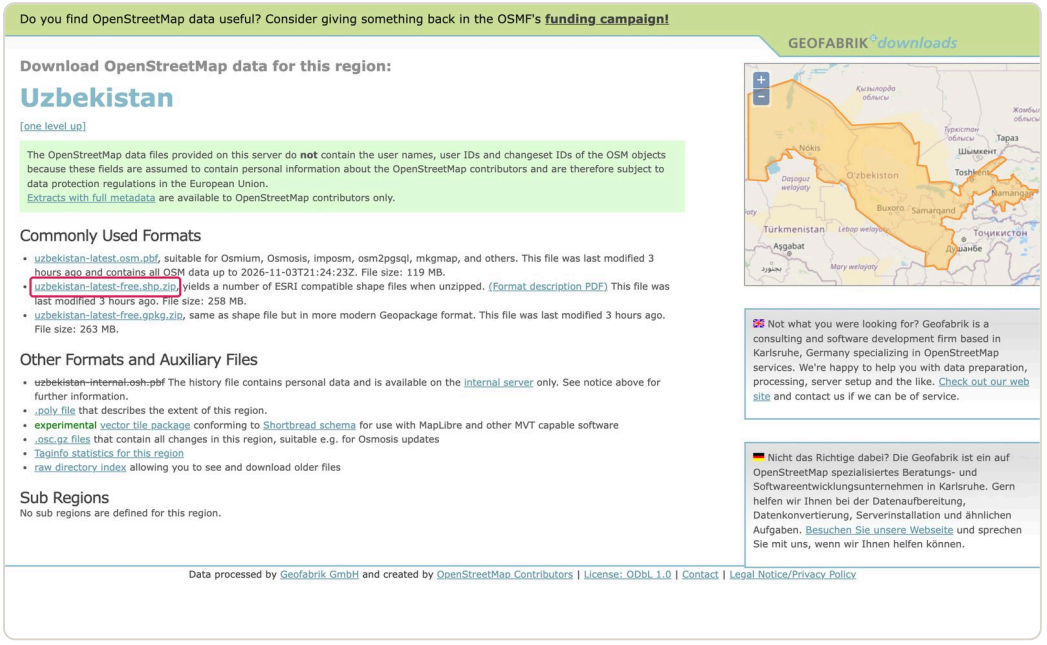
Nima kerak	Yordamchiga nima yoziladi
kafelar	`amenity=cafe`
maktablar	`amenity=school`
dorixonalar	`amenity=pharmacy`
masjidlar	`amenity=place_of_worship and religion=muslim`
avtobus bekatlari	`highway=bus_stop`
bog‘lar	`leisure=park`
mehmonxonalar	`tourism=hotel`
supermarketlar	`shop=supermarket`

So‘ng `in` va shahar nomini lotincha yozing: `in Samarkand`, `in Bukhara`, `in Namangan`. Boshqa narsaning tegini topish uchun OSM vikidagi Map features sahifasini ko‘ring yoki openstreetmap.org da **?** (Obyektlar haqida ma’lumot) vositasi bilan joyni bosing.

> 💡 Agar Overpass Turbo server band yoki so‘rov vaqti tugadi desa, bir daqiqa kutib, yana **Run** ni bosing yoki kichikroq shaharni tanlang.

B · Geofabrik (butun mamlakat)

1. **download.geofabrik.de/asia/uzbekistan.html** ni oching va **uzbekistan-latest-free.shp.zip** ni bosing.



2. Arxivni oching. Ichida shapefile formatidagi ko‘plab qatlamlar bor: `gis_osm_roads_free_1` (yo‘llar), `gis_osm_waterways_free_1` (daryolar), `gis_osm_buildings_a_free_1` (binolar), `gis_osm_pois_free_1` (qiziqarli joylar: kafelar, do‘konlar, maktablar…) va boshqalar.
3. `gis_osm_pois_free_1.shp` ni QGIS’ga sudrab tashlang. `fclass` ustuni har bir nuqta nima ekanini bildiradi.

4. Faqat kafe, restoran va fastfudni qoldirish uchun: qatlam ustida o‘ng tugma → **Filter…** va quyidagini yozing:

```
"fclass" IN ('cafe', 'restaurant', 'fast_food')
```

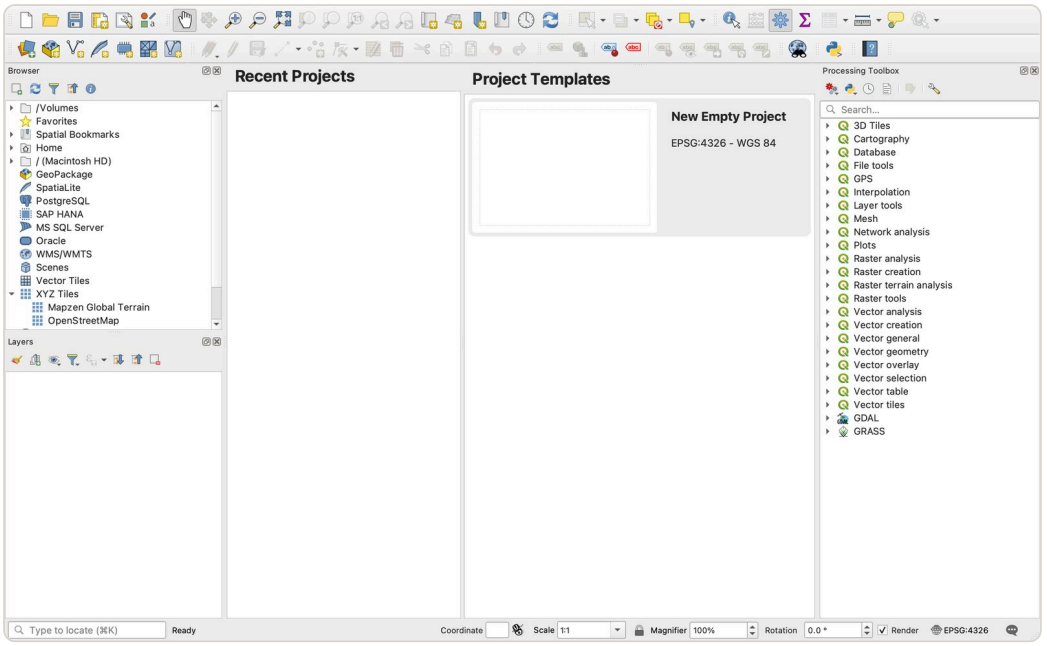
5. Faqat Toshkentni qoldirish uchun: **Vector** → **Geoprocessing Tools** → **Clip** (Kesish), bunda **Input layer** nuqtalar, **Overlay layer** esa `tashkent_districts`.

Ushbu dars fayli aynan shunday tayyorlangan (OSM’da ba’zi joylar bino sifatida chizilgan, shuning uchun ular ham `gis_osm_pois_a_free_1` qatlamidan nuqtaga aylantirilgan).

> ⚖️ **Har doim manbani ko‘rsating:** © *OpenStreetMap contributors*. OSM’dan ODbL litsenziyasi asosida bepul foydalanish mumkin, faqat ma’lumot qayerdan olinganini yozish kerak.

🖥️ 3-qism · QGIS bilan tanishuv

QGIS’ni oching. Boshlang‘ich oynani ko‘rasiz:

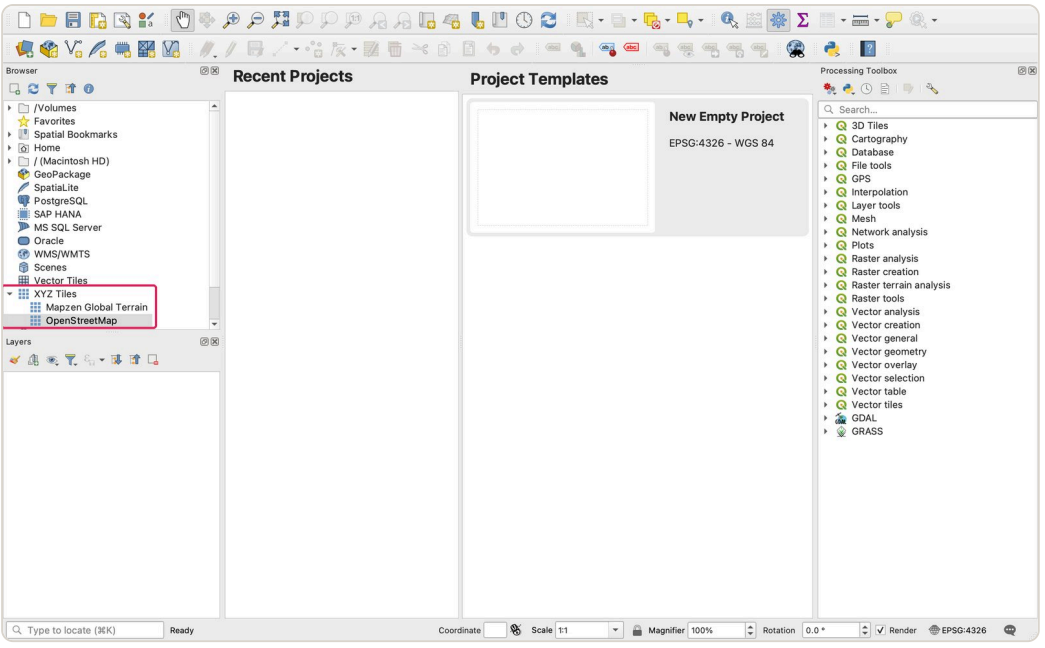


Sizga kerak bo‘ladigan to‘rt qism:

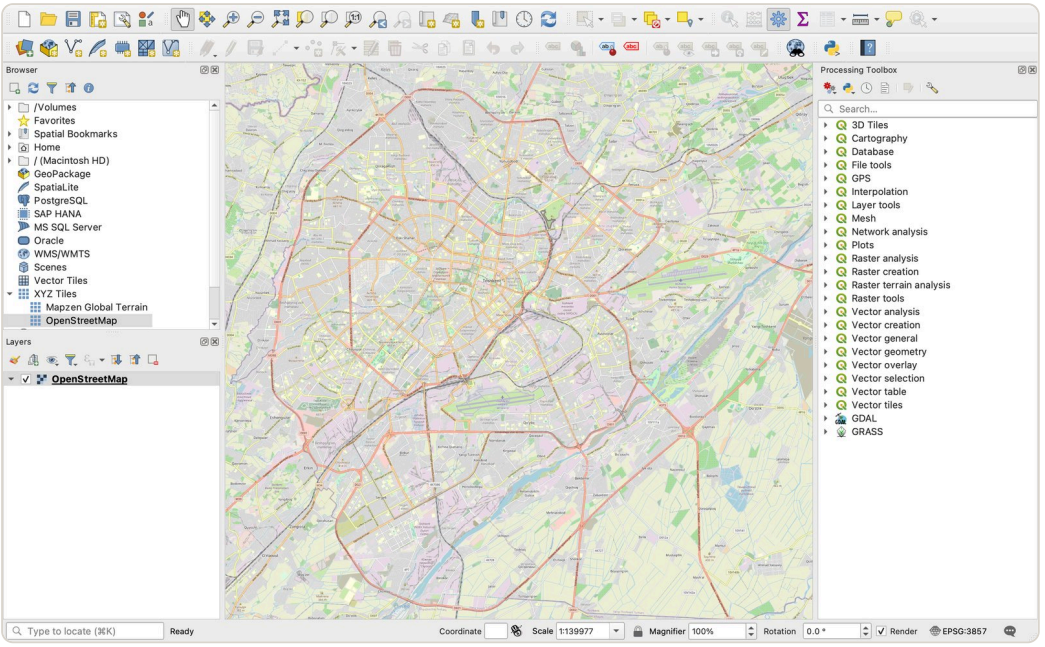
- **Browser** (Brauzer, chap yuqorida): fayllaringiz va onlayn xaritalar.
- **Layers** (Qatlamlar, chap pastda): xaritadagi hamma narsa. Yuqoridagi qatlamlar pastdagilarning ustidan chiziladi.
- **Xarita** (markazda). Sichqoncha g‘ildiragi yaqinlashtiradi, tugmani bosib turib tortsangiz, xarita suriladi.
- **Processing Toolbox** (Tahlil vositalari paneli, o‘ngda): yuzlab vositalar va qidiruv qatori. Agar ko‘rinmasa, **Processing** → **Toolbox** ni oching.

🗺️ 4-qism · Fon xaritasini qo‘shing

Browser panelida **XYZ Tiles** yonidagi kichik strelkani bosing, so‘ng **OpenStreetMap** ni **ikki marta bosing**.

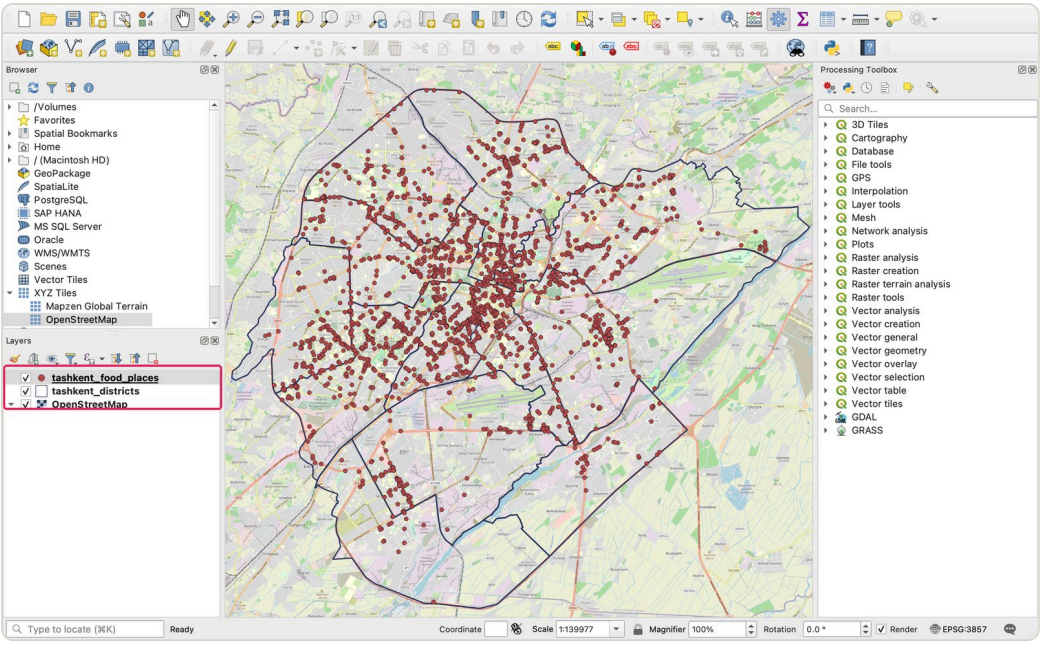


Dunyo xaritasi paydo bo‘ladi. Toshkentni sichqoncha g‘ildiragi bilan yaqinlashtiring yoki 5-qismni kuting: tumanlar yuklangach, `tashkent_districts` ustida sichqonchaning o‘ng tugmasini bosing → **Zoom to Layer** (Qatlamga yaqinlashtirish), xarita darhol o‘sha yerga o‘tadi.

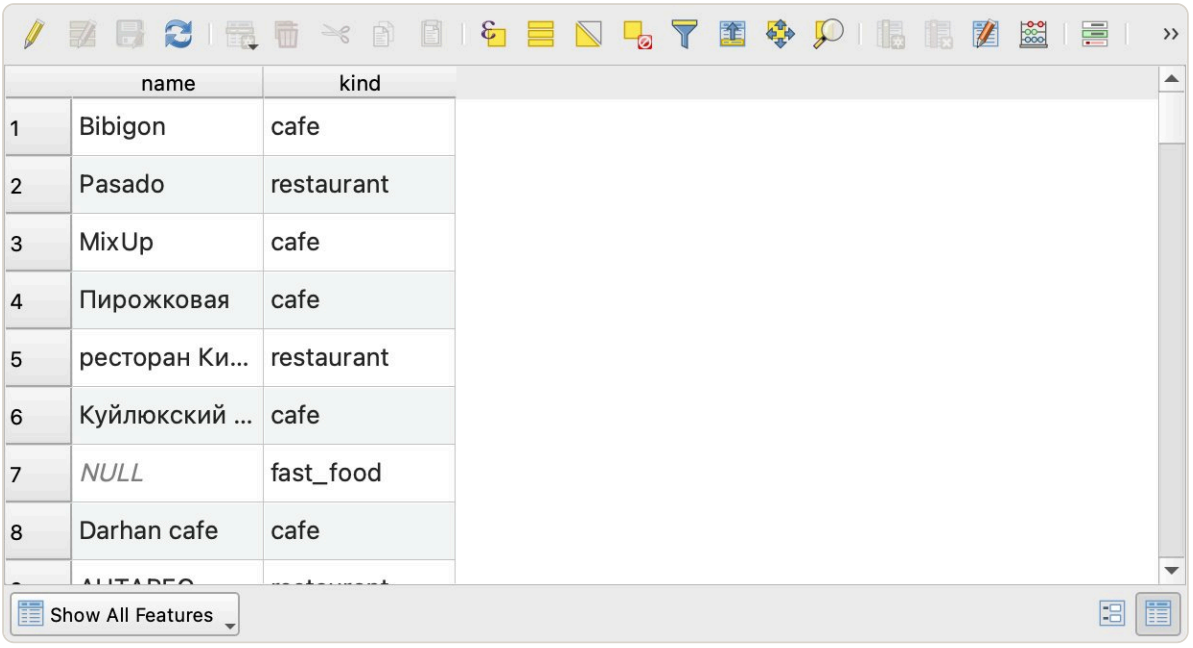


📍 5-qism · Joylar va tumanlarni qo‘shing

1. Papkangizdan QGIS xaritasiga avval `tashkent_districts.geojson` ni, so‘ng `tashkent_food_places.geojson` ni sudrab tashlang. Ular **Layers** panelida paydo bo‘ladi.



2. Ichiga qarang: `tashkent_food_places` ustida o‘ng tugmani bosing → **Open Attribute Table** (Atributlar jadvalini ochish). Har bir qator bitta joy: nomi `name` va turi `kind` (`cafe`, `restaurant` yoki `fast_food`). Ba’zilarining nomi yo‘q (`NULL`): uni hali hech kim OpenStreetMap’ga qo‘shmagan.



	name	kind
1	Bibigon	cafe
2	Pasado	restaurant
3	MixUp	cafe
4	Пирожковая	cafe
5	ресторан Ки...	restaurant
6	Куйлюкский ...	cafe
7	*NULL*	fast_food
8	Darhan cafe	cafe

Show All Features

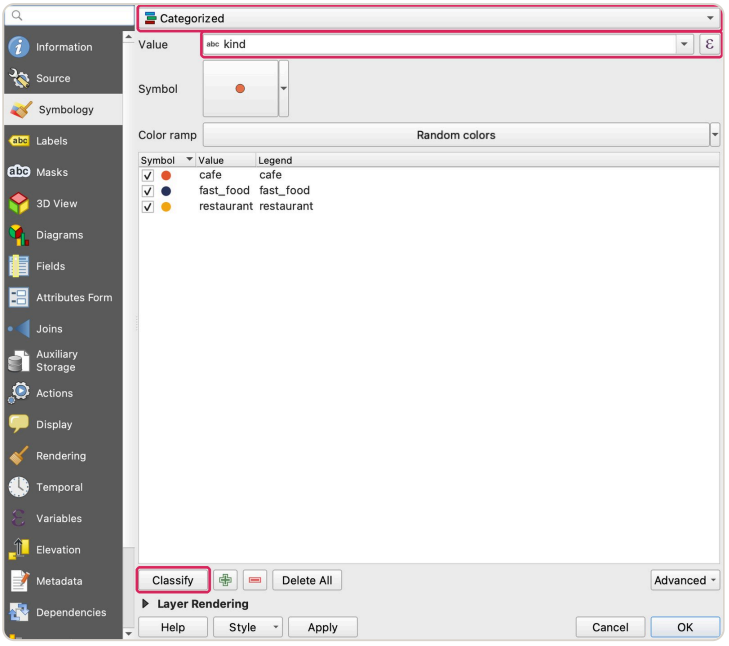
3. Jadvalni yoping.

> 💡 Agar tumanlar xaritani bir xil rangga bo‘yab qo‘ysa: `tashkent_districts` ni ikki marta bosing → **Symbology** → **Simple Fill** ni bosing → **Fill style** da *No Brush* ni tanlang. Endi faqat chegaralar ko‘rinadi.

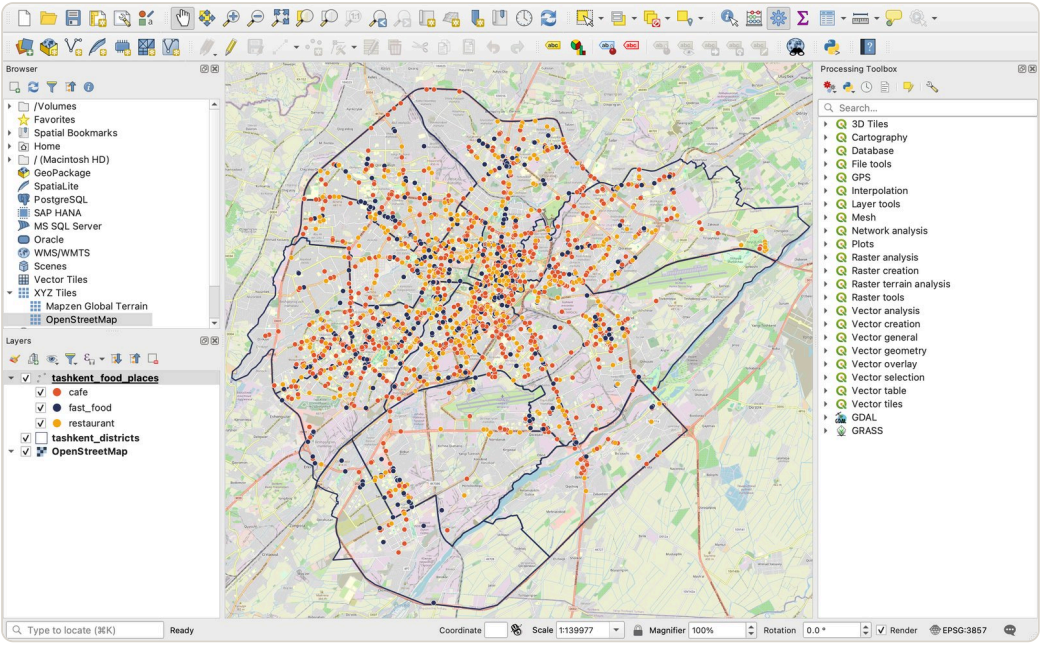
🎨 6-qism · Joylarni turiga qarab bo‘yang

1. Layers panelida `tashkent_food_places` ni **ikki marta bosing**. **Layer Properties** (Qatlam xususiyatlari) oynasi ochiladi.

2. Chapda **Symbology** (Uslub) ni bosing.
3. Yuqorida **Single Symbol** ni **Categorized** (Toifalar bo‘yicha) ga almashtiring.
4. **Value** maydonida `kind` ni tanlang.
5. **Classify** ni, so‘ng **OK** ni bosing.

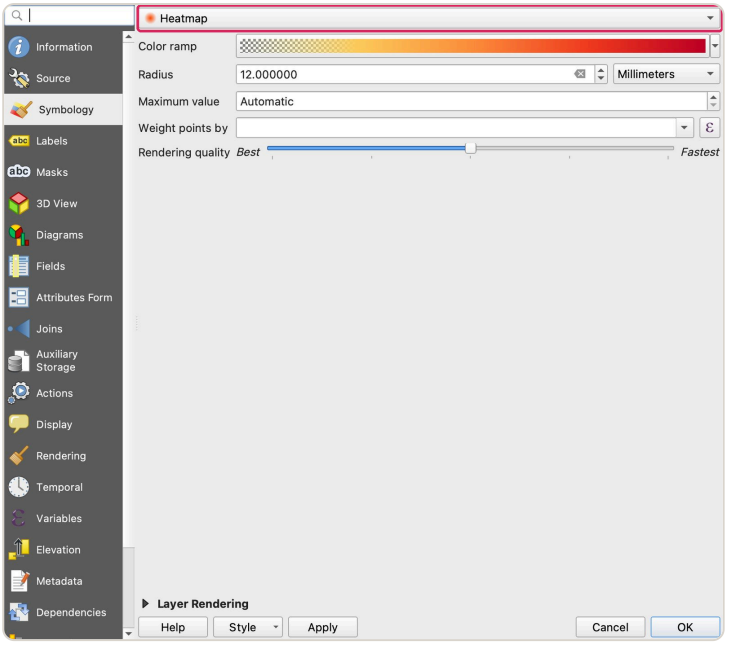


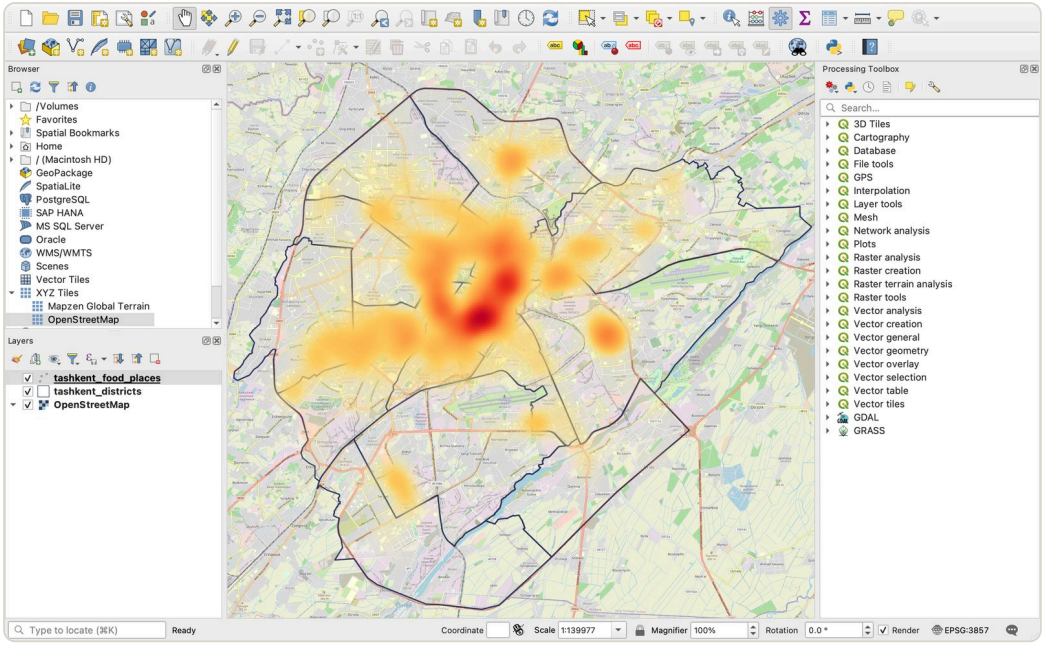
Endi kafe, restoran va fastfudning o‘z ranglari bor:



🔥 7-qism · Issiqlik xaritasi: qaynoq nuqtalarni ko‘ring

1. `tashkent_food_places` ni yana ikki marta bosing → **Symbology**.
2. **Categorized** ni **Heatmap** (Issiqlik xaritasi) ga almashtiring.
3. Yoqqan **Color ramp** (Rang shkalasi) ni tanlang (yonidagi strelka, masalan *Reds* yoki *Magma*) va **Radius** ni taxminan `12` millimetr qiling. **OK** ni bosing.





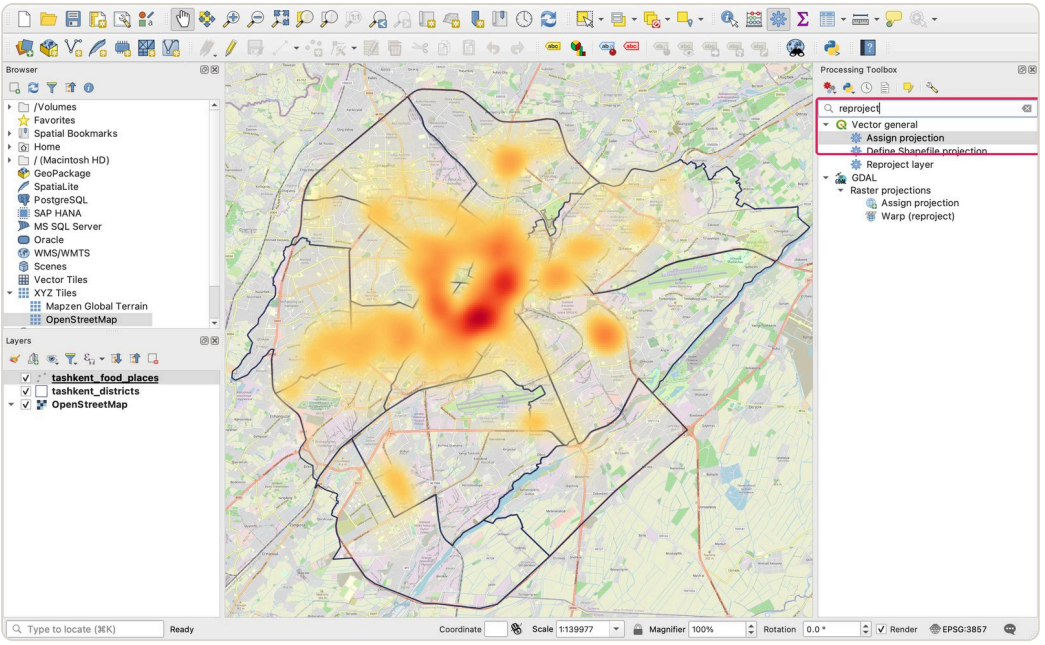
Eng qaynoq joy **shahar markazi**. Kichikroq qaynoq nuqtalar katta ko‘chalar bo‘ylab va mahallalarda ko‘rinadi.

> 💡 Fon xaritasi chalg‘itmasligi uchun: `OpenStreetMap` ni ikki marta bosing → **Transparency** → **Global Opacity** ni taxminan 50% qiling.

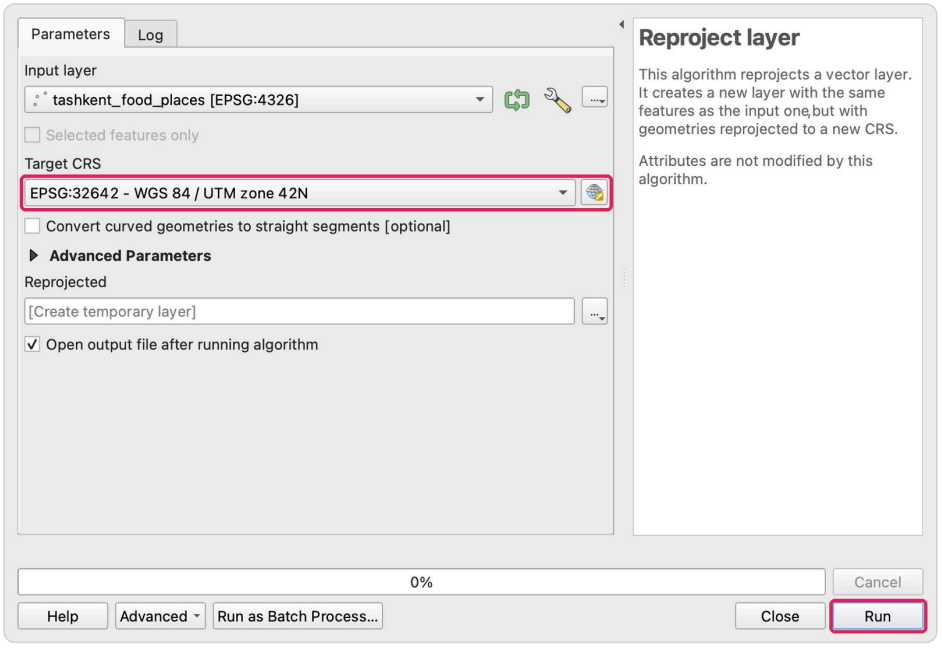
📏 8-qism · Gradusdan metrga (qayta proyeksiyalash)

DBSCAN masofani qatlam birliklarida o‘lchaydi. Nuqtalarimiz **gradusda**, demak "300" deganda 300 gradus, ya’ni deyarli butun yer shari tushuniladi! Ularni **metrga** o‘tkazamiz.

1. **Processing Toolbox** qidiruv qatoriga `reproject` deb yozing va **Reproject layer** (Qatlamni qayta proyeksiyalash) ni ikki marta bosing.

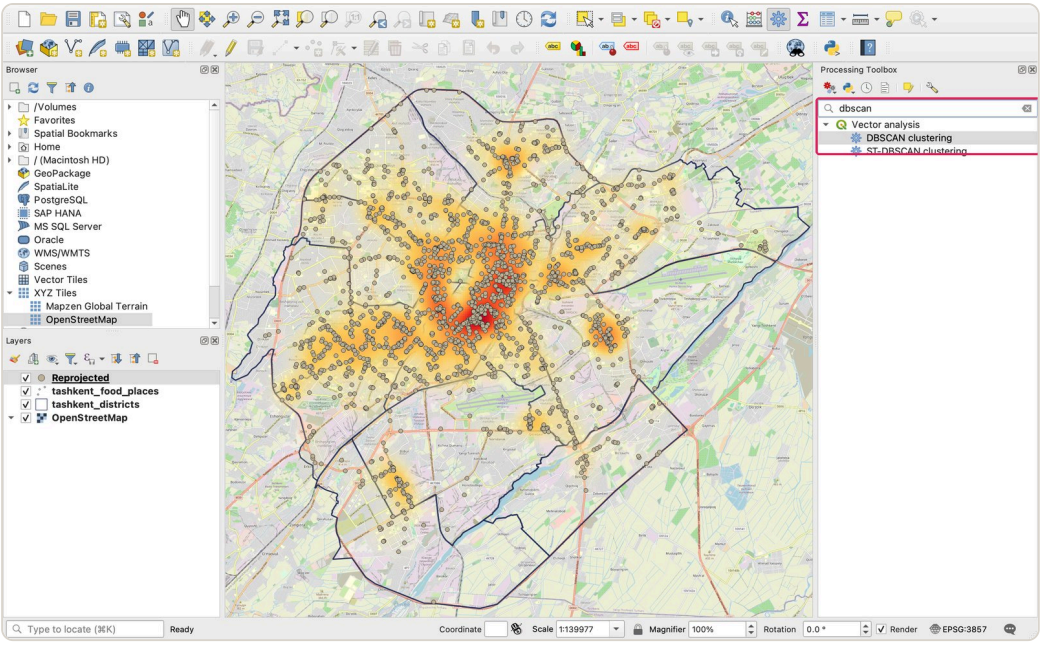


2. **Input layer:** `tashkent_food_places`.
3. **Target CRS:** globus belgili kichik tugmani bosing va `32642` ni qidiring. **WGS 84 / UTM zone 42N** (EPSG:32642) ni tanlang: bu Toshkent uchun metrli koordinatalar tizimi.
4. **Run** ni, so‘ng **Close** ni bosing. **Reprojected** nomli yangi qatlam paydo bo‘ladi.

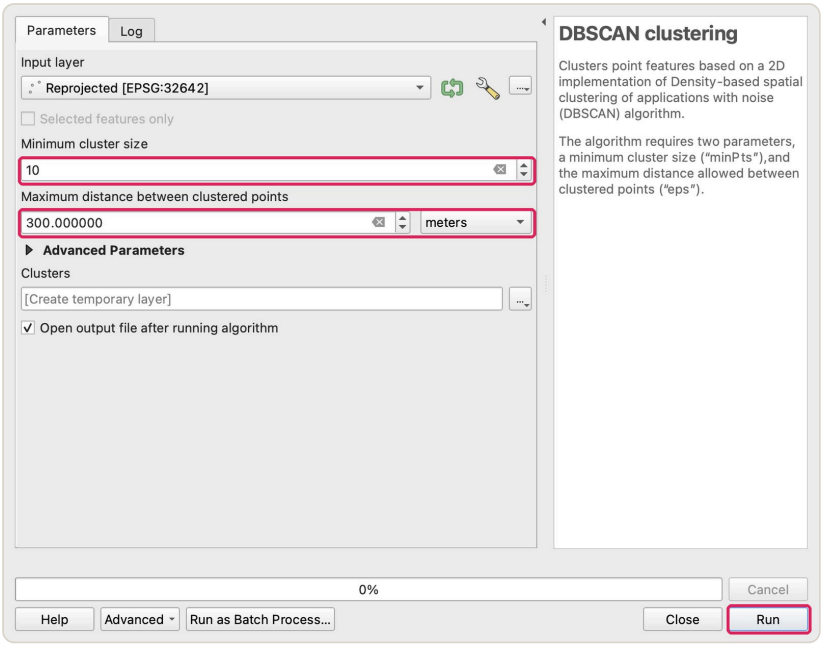


9-qism · Klasterlarni toping (DBSCAN)

1. Processing Toolbox’da `dbscan` ni qidiring va **DBSCAN clustering** ni ikki marta bosing.

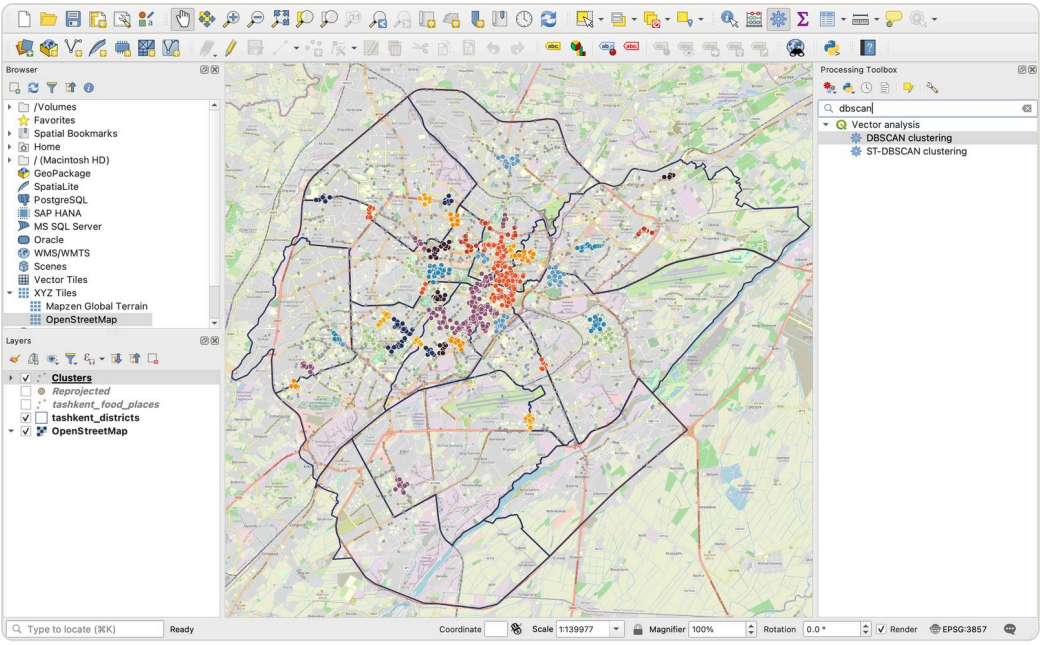


2. To‘ldiring:
    - **Input layer:** `Reprojected`
    - **Minimum cluster size** (Klasterning eng kichik o‘lchami): `10`
    - **Maximum distance between clustered points** (Klaster nuqtalari orasidagi eng katta masofa): `300` (meters)
3. **Run** ni, so‘ng **Close** ni bosing.



Clusters nomli yangi qatlam paydo bo‘ladi. Uning atributlar jadvalini oching: ikkita yangi ustun bor, `CLUSTER_ID` (joy qaysi klasterga kiradi) va `CLUSTER_SIZE` (shu klasterda nechta joy bor). `NULL` bo‘lgan joylar shovqin: ular hech qaysi klasterga kirmaydi.

4. Uni bo‘yang: **Clusters** ni ikki marta bosing → **Symbology** → **Categorized** → **Value:** `CLUSTER_ID` → **Color ramp:** *Random colors* → **Classify** → **OK**. Boshqa nuqtali qatlamlarni yashirish uchun ularning belgisini olib tashlang.

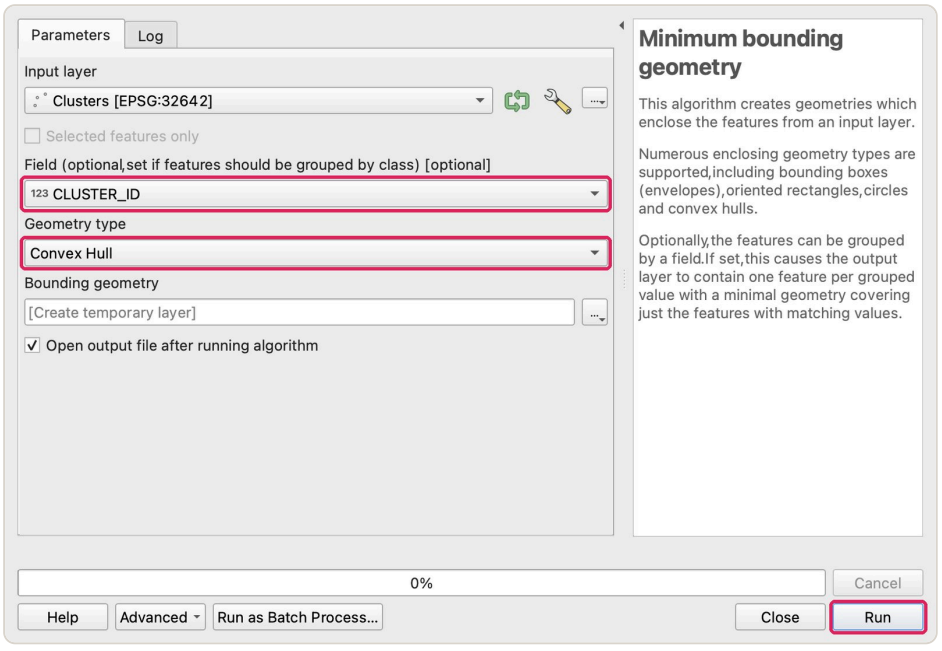


QGIS **46 ta klaster** topdi. Ularda jami **2 408 ta joydan 1 159 tasi** bor (deyarli yarmi!). Eng katta klasterda **243 ta joy** bor: u shahar markazida, **Mirobod va Yakkasaroy** tumanlari tutashgan joyda.

> 🧪 **Tajriba:** DBSCAN'ni `200` yoki `500` metr bilan qayta ishga tushiring. 200 m da klasterlar kichikroq va zichroq chiqadi, 500 m da esa markaz mingdan ortiq joydan iborat bitta ulkan klasterga aylanadi. Yagona to‘g‘ri javob yo‘q: hikoyangizni eng yaxshi aytib beradiganini tanlang.

🔷 10-qism · Har bir klaster atrofiga shakl chizing

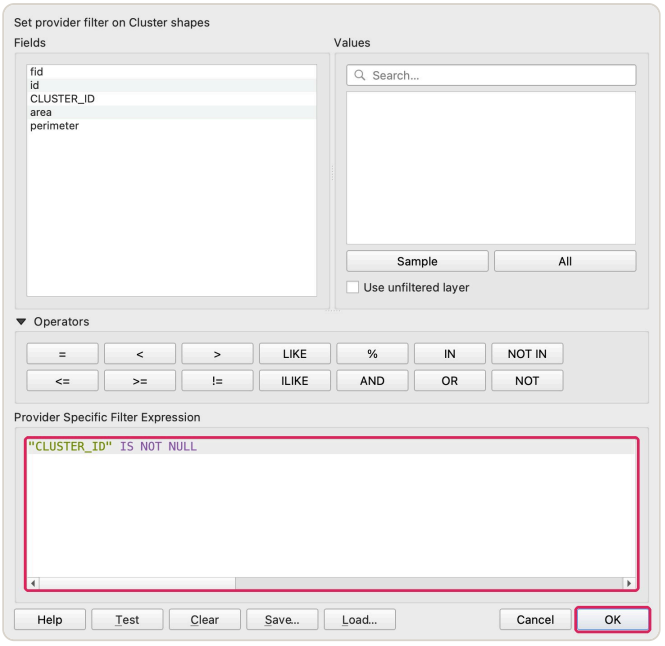
1. Processing Toolbox'da `minimum bounding` ni qidiring va **Minimum bounding geometry** (Eng kichik chegaralovchi geometriya) ni ikki marta bosing.
2. **Input layer:** `Clusters` · **Field:** `CLUSTER_ID` · **Geometry type: Convex Hull** (Qavariq qobiq). **Run** ni, so‘ng **Close** ni bosing.



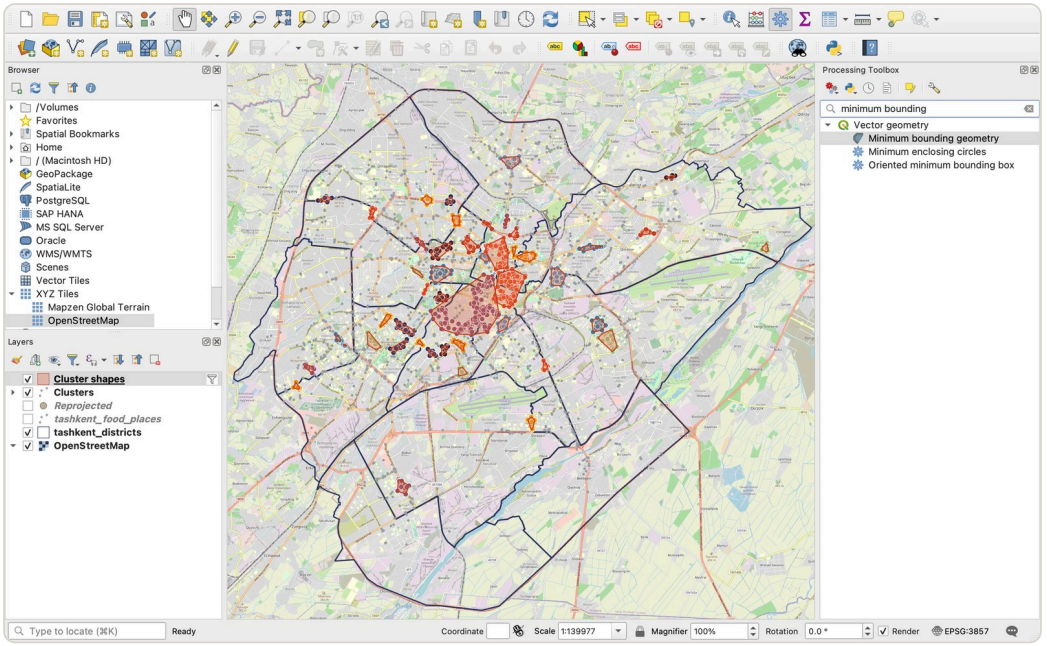
Qavariq qobiq har bir klaster nuqtalari atrofiga tortilgan rezinkaga o‘xshaydi. **Bounding geometry** nomli yangi qatlam paydo bo‘ladi (biz uni *Cluster shapes* deb qayta nomladik: o‘ng tugma → **Rename Layer**).

3. Shovqin nuqtalari (klastersiz) ham bitta katta shakl oldi. Uni olib tashlang: yangi qatlam ustida o‘ng tugmani bosing → **Filter…** (Filtr), quyidagi ifodani yozing va **OK** ni bosing:

```
"CLUSTER_ID" IS NOT NULL
```



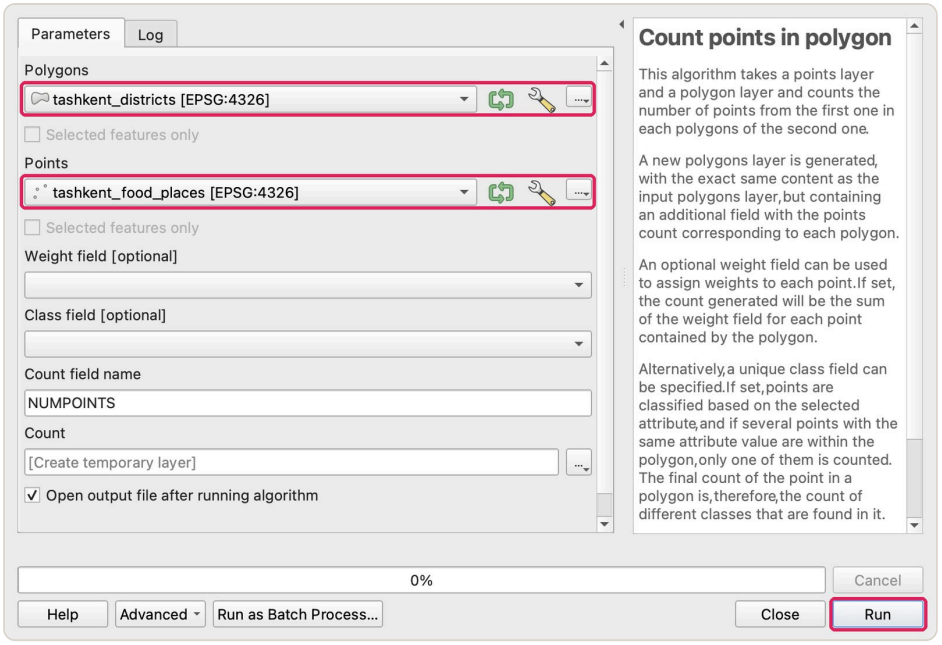
4. Shakllarni shaffof qiling: qatlamni ikki marta bosing → **Symbology** → bo‘yoq rangining **shaffofligini** (opacity) taxminan 30% qiling.



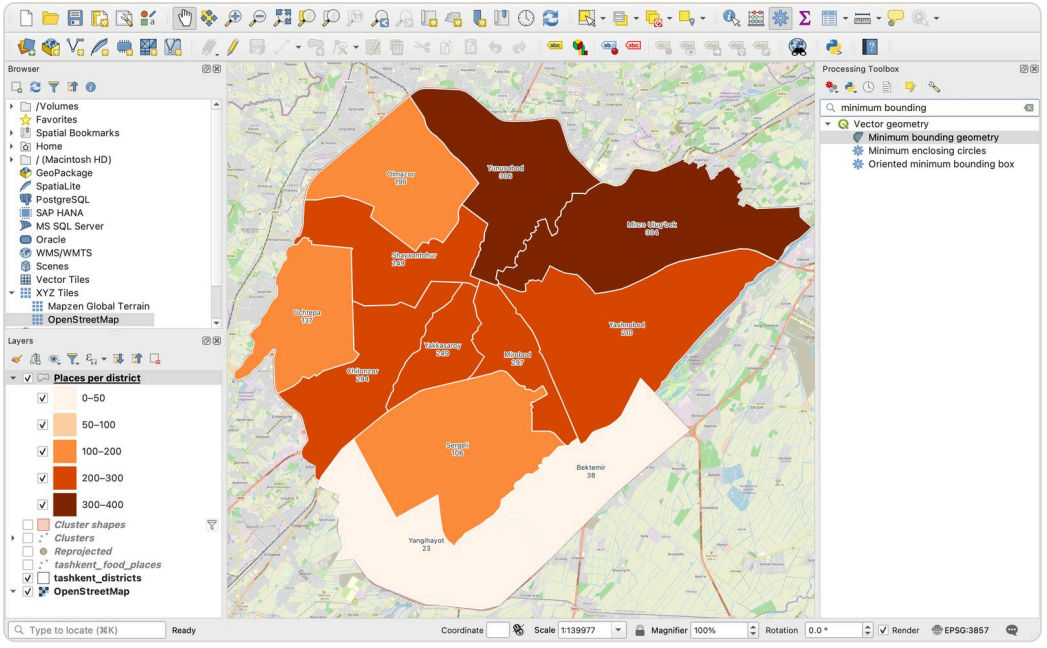
🏙️ 11-qism · Har bir tumanda nechta joy bor?

1. Menyuda **Vector** → **Analysis Tools** → **Count Points in Polygon** (Poligondagi nuqtalarni sanash) ni oching yoki Toolbox’da `count points` ni qidiring.

2. **Polygons:** `tashkent_districts` · **Points:** `tashkent_food_places`. **Run** ni, so‘ng **Close** ni bosing.



Yangi **Count** qatlamida `NUMPOINTS` ustuni bor. Uni bo‘yang: **Symbology** → **Graduated** (Darajali) → **Value:** `NUMPOINTS` → rang shkalasini tanlang → **Classify** → **OK**. Yozuvlar qo‘shing: **Labels** → **Single Labels** → **Value:** `district`.

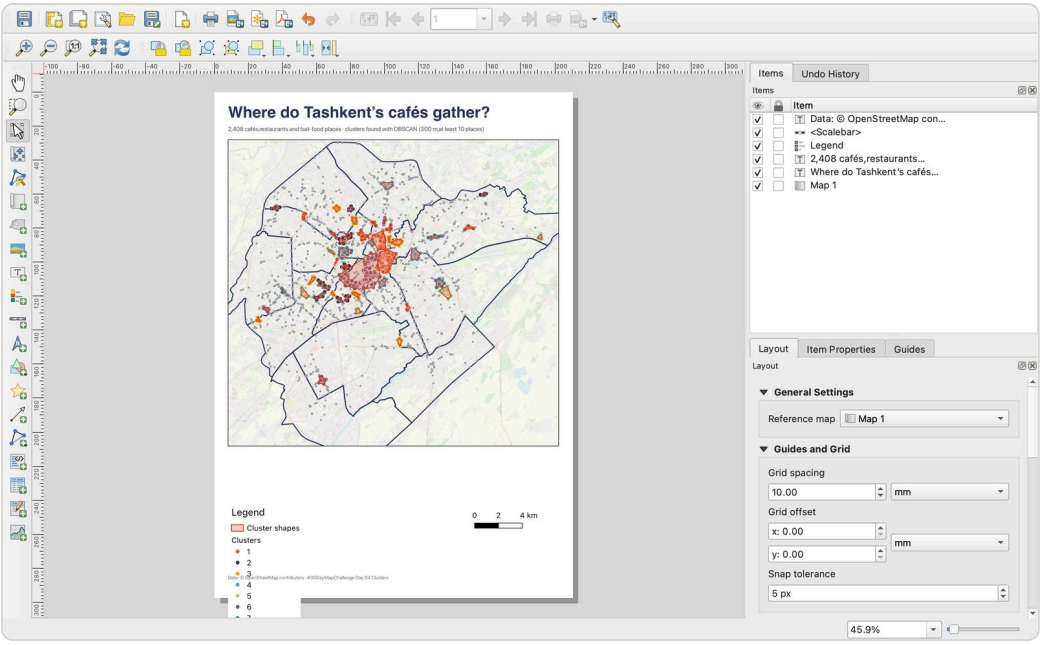


Tuman	Joylar	Maydon (km²)	1 km² dagi joylar
Mirobod	297	16,6	17,9
Yakkasaroy	249	14,0	17,8
Chilonzor	294	30,7	9,6
Shayxontohur	249	27,6	9,0
Yunusobod	306	40,5	7,5
Olmazor	195	33,0	5,9
Mirzo Ulug‘bek	304	55,6	5,5
Uchtepa	137	28,3	4,8
Yashnobod	210	59,1	3,6
Sergeli	106	51,9	2,0
Bektemir	38	31,9	1,2
Yangihayot	23	44,5	0,5

Eng ko‘p joy **Yunusobod** tumanida, lekin eng zich tumanlar markazdagi kichik **Mirobod** va **Yakkasaroy** tumanlari: har bir km² da deyarli 18 ta joy.

🖨️ 12-qism · Chop etishga tayyor xarita yasang (Print Layout)

1. **Project** → **New Print Layout** (Yangi maket), nom bering (masalan, `Clusters map`) va **OK** ni bosing. Bo‘sh varaqli yangi oyna ochiladi.
2. **Add Item** → **Add Map** ni bosing va varaqda to‘rtburchak chizing. Ichida xaritangiz paydo bo‘ladi.
3. Sarlavha uchun **Add Item** → **Add Label** (matn o‘ngdagi **Item Properties** da yoziladi, shriftni kattaroq qiling).
4. **Add Item** → **Add Legend** va **Add Item** → **Add Scale Bar** ni qo‘shing va ularni xarita ostiga joylashtiring.
5. **Layout** → **Export as Image…** ni bosing, nom bering va PNG sifatida saqlang.



💡 Keraksiz qatlamlarni (masalan, 46 ta klaster raqamini) olib tashlash uchun legendaning **Item Properties** bo‘limida **Auto update** belgisini olib tashlang. Har doim manbani yozing: *Data © OpenStreetMap contributors*.

🌈 Xaritani o‘zingizniki qiling: g‘oyalar

- **Boshqa joylar:** xuddi shu qadamlar har qanday nuqtalar uchun ishlaydi. Maktablar, dorixonalar yoki supermarketlarni o‘zingiz yuklab oling (yuqoridagi "Ma’lumotlar qayerdan olingan?" bo‘limi) yoki 01-kun darsidagi yong‘in nuqtalaridan foydalaning.
- **Faqat kafelar:** qatlam ustida o‘ng tugma → **Filter…** → `"kind" = 'cafe'`, so‘ng DBSCAN’ni qayta ishga tushiring.
- **Tungi rejim:** OpenStreetMap qatlamini olib tashlang, fonni to‘q rangga o‘zgartiring (**Project** → **Properties** → **General** → **Background color**) va issiqlik xaritasi uchun *Magma* kabi yorqin shkalani tanlang.
- **O‘z shahringiz:** Samarqand, Buxoro yoki Namangan? Overpass Turbo Yordamchisida `in Samarkand` deb yozing va qadamlarni takrorlang. 8-qismda shahringiz uchun UTM zonasini tanlang: Toshkent, Samarqand, Namangan va Farg‘ona uchun **EPSG:32642**; Buxoro, Navoiy va Urganch uchun **EPSG:32641**; Nukus uchun **EPSG:32640**; Andijon uchun **EPSG:32643**.

💾 Xaritangizni ulashing

1. Xaritangiz **mavzusi** nomi bilan papka yarating, masalan `Coffee clusters of Tashkent`. Ichiga xaritangizni va **ismingiz**, ma’lumotlar (OpenStreetMap) va vosita (QGIS) yozilgan qisqa `README.md` faylini qo‘ying.
2. Papkani ushbu repozitoriydagi `2026/Day 04 Clusters/` papkasiga yuklang. Kodsiz qanday qilish asosiy README’dagi Quick Start bo‘limida yozilgan.

3. Xaritani **#30DayMapChallenge** heshtegi va **© OpenStreetMap contributors** yozuvi bilan e’lon qiling.

🆘 Nimadir ishlamadimi?

Nimani ko‘ryapsiz	Nima qilish kerak
Processing Toolbox ko‘rinmayapti	**Processing** → **Toolbox** (yoki **View** → **Panels** → **Processing Toolbox**).
DBSCAN **0 ta klaster** yoki bitta ulkan klaster topyapti	Siz gradusdagi qatlamni oldingiz. DBSCAN’ni **Reprojected** qatlamida ishga tushiring (8-qism) va masofa birligi *meters* ekanini tekshiring.
Tumanlar nuqtalarni yopib qo‘yyapti	Layers panelida `tashkent_districts` ni nuqtali qatlamlardan **pastga** suring yoki bo‘yoqni *No Brush* qiling (5-qism).
OpenStreetMap fon xaritasi kulrangligicha qolyapti	Internet kerak. Ulanishni tekshiring va xaritani qayta yuklash uchun biroz yaqinlashtiring.
Vosita qatlam noto‘g‘ri deyapti	Vosita yuqorisidagi **Input layer** ro‘yxatida to‘g‘ri qatlam tanlanganini tekshiring.
macOS QGIS’ni ochib bo‘lmaydi deyapti	*Applications* papkasida QGIS ustida o‘ng tugmani bosing → **Open** → **Open**. Buni faqat bir marta qilish kerak.
Qatlamim yo‘qoldi	Layers panelida uning yonidagi ✓ belgini tekshiring, so‘ng o‘ng tugma → **Zoom to Layer**.

📎 Manbalar

- Kafe, restoran va fastfud joylari: © OpenStreetMap ishtirokchilari (ODbL), Geofabrik’ning 2026-yil 2-noyabrdagi yuklamasi. `data/README.md` ga qarang.
- Toshkent tumanlari: ushbu repozitoriyning `Boundaries` papkasidan.
- Vosita: QGIS, bepul va ochiq kodli. DBSCAN: Ester, Kriegel, Sander va Xu (1996).